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AFTERHOURS 1.0

CODE BEYOND THE CLOCK

TraceRoot

Graph-Causal Incident Investigation & Autonomous Containment Platform

Problem Statement 01: Outage Investigator | Network Incident Root-Cause Analysis

TEAM DETAILS

Team Name: CloudGuardians

Domain: Cybersecurity · Network Observability · SRE

TEAM MEMBERS

- Lead / Architecture: Yashwanth KR
- Ingestion & Causal Graph: Team Member 2
- Incident Engine & Playbooks: Team Member 3
- Dashboard & Observability UI: Team Member 4

Problem Statement & Its Relevance

PS 01: Outage Investigator | Navigating the Fog of War in Distributed Incidents

WHAT EXACT PROBLEM ARE WE SOLVING?

- **The 'Alert Storm' Paradox:** A single root failure (e.g., BGP route leak, DNS timeout, DB pool exhaustion) triggers thousands of secondary dependent alerts across microservices.
- **Symptom vs. Cause Confusion:** Traditional monitoring treats all failing services identically. On-call engineers cannot isolate what started the collapse from downstream victims.
- **High Cognitive Triage Load:** Incident responders spend crucial early minutes manually digging through unranked logs rather than executing containment.

FOR WHOM IS THIS A PROBLEM?

- **Site Reliability Engineers (SREs):** Engineers under severe pressure to restore mission-critical services under strict 99.99% SLAs.
- **Network Operations Centers (NOCs):** Operators managing multi-tier hybrid/cloud topologies lacking end-to-end visibility across virtual & physical layers.
- **Enterprise Cloud Architects:** Teams designing microservices where complex cascading dependency graphs obscure failure propagation.
- **Digital Businesses & End Users:** Fintechs, e-commerce, and healthcare where every minute of outage erodes user trust and halts operations.

WHY DOES THIS PROBLEM MATTER?

- **Massive Financial Bleed:** Enterprise downtime costs exceed $9,000 per minute ($540,000/hour) according to Gartner and Ponemon studies.
- **Prolonged MTTI / MTTR:** Root-cause identification consumes >70% of total outage triage time due to manual trial-and-error hypothesis testing.
- **Cascading Blast-Radius:** Uncontained network or service failures quickly spread to healthy availability zones, escalating localized faults into full outages.
- **Operator Burnout & Fatigue:** Repeated false-alarm storms and chaotic firefighting lead to human errors during live sev-1 remediation.

Proposed Solution & Innovation

TraceRoot: Temporal-Topological Causal Graph Engine for Rapid Containment

END-TO-END INVESTIGATION & REMEDIATION PIPELINE

[1] Multi-Source Ingestion ➔ [2] Topological Dependency Graph ➔ [3] Temporal-Causal Engine ➔ [4] Actionable Containment

CORE SYSTEM CAPABILITIES

- **Multi-Modal Telemetry Stream:** Ingests simulated VPC flow logs, packet metrics, application stderr logs, and Kubernetes events in real time.
- **Dynamic Causal DAG:** Constructs a directed acyclic graph mapping services, subnets, routers, and DB clusters to model real dependency chains.
- **Temporal Fault Sequencing:** Traces exact timestamped event propagation (T_0 root vs. $T_0 + \Delta t$ symptoms) to identify the initial point of failure.
- **Incident Timeline Reconstruction:** Generates an interactive, chronological Gantt view of the incident from initial anomaly to full blast radius.

INNOVATIVE APPROACH & HIGHLIGHTS

- **Deterministic Causality vs. Guesswork:** Uses graph reachability algorithms (Floyd-Warshall / PageRank) instead of black-box statistical correlation.
- **Explainable Evidence Dossier:** Every flagged root cause is paired with concrete cryptographic/log receipts and dependency paths (zero hallucination).
- **Closed-Loop Containment Playbooks:** Recommends validated containment actions (dynamic DNS reroute, subnet isolation, rate limiting) with human-in-the-loop approvals.
- **Noise-Suppression Intelligence:** Automatically de-duplicates cascading symptom alerts into a single unified incident graph.

Existing Solutions & Differentiation

Why Conventional APMs and GenAI Fall Short in Outage Investigation

COMPARISON: TRACEROOT VS. EXISTING INDUSTRY TOOLS

- **1. Root-Cause Isolation Paradigm**
  - ❌ Traditional APM (Datadog/Dynatrace): Relies on static metric thresholds and statistical co-occurrence.
  - ✅ **TraceRoot Advantage: Deterministic graph reachability and temporal causality isolate the true source with proof.**
- **2. Cascading Failure Handling**
  - ❌ Standard Monitoring & Alerting: Floods operators with hundreds of redundant symptom alarms ('Alert Fatigue').
  - ✅ **TraceRoot Advantage: Group-collapses downstream symptom cascades into a single unified causal tree.**
- **3. Explainability & Trust**
  - ❌ GenAI Log Copilots: Prone to hallucinations, lacks architectural topology, and offers non-verifiable suggestions.
  - ✅ **TraceRoot Advantage: 100% explainable, deterministic evidence trail linking physical logs to graph topology.**
- **4. Incident Containment & Action**
  - ❌ Existing Observability Suites: Passive observability dashboards that leave containment entirely to manual triage.
  - ✅ **TraceRoot Advantage: Generates ready-to-execute, guardrailed network & cloud containment playbooks**

OUR KEY COMPETITIVE ADVANTAGE: MATHEMATICALLY DETERMINISTIC CAUSALITY

Unlike probabilistic ML models that guess correlations or black-box LLMs that hallucinate incident causes, TraceRoot merges physical network topology with nanosecond event ordering. SREs receive concrete proof, confidence metrics, and targeted containment actions in <60 seconds.

Technical Approach & Feasibility

Full-Stack Architecture, Algorithmic Engine & 24-Hour Hackathon Build Scope

SYSTEM ARCHITECTURE

- **Ingestion & Normalizer:** Python FastAPI & Apache Kafka stream processing; normalizes OpenTelemetry logs, VPC flow metrics, and ping health stats.
- **Graph Topology Engine:** NetworkX & Neo4j graph database storing network nodes, routing hops, microservice edges, and upstream dependencies.
- **Causal Graph Traversal:** Custom Dijkstra/Floyd-Warshall fault path algorithms and PageRank centrality to isolate failure epicenters.
- **Incident Command UI:** React / Next.js frontend with Cytoscape.js interactive topology graph and Chrono-Timeline view.

ROLE OF AI & DETERMINISTIC RULES

- **Deterministic Graph Rules (Core):** Root cause attribution relies strictly on graph reachability and temporal ordering, guaranteeing 100% explainability and zero hallucination.
- **Targeted NLP (Optional / Supporting):** Sentence-BERT embeddings cluster noisy, unstructured error logs into standardized canonical error tokens without manual regex.
- **Dynamic Anomaly Detection:** Z-score sliding window on latency and packet-loss metrics flags pre-outage degradation before total service failure.
- **Human-in-the-Loop Safeguards:** Remediation commands require operator approval; no automated catastrophic actions.

HACKATHON BUILD SCOPE

- **Synthetic Chaos Generator:** Simulates 4 realistic outage scenarios: BGP route drop, DNS resolver exhaustion, DB deadlocks, and cascading API timeouts.
- **Real-Time Ingestion Pipeline:** Fully functional streaming pipeline processing simulated multi-source logs at 500+ events/sec.
- **Live Interactive Dashboard:** Interactive Cytoscape.js topology view showing nodes changing status from healthy to degraded, pinpointing root cause in red.
- **Remediation Playbook Generator:** Outputs ready-to-run AWS CLI / iptables containment commands for the identified fault.

Expected Impact & Execution Plan

Quantifiable SRE Metrics, 24-Hour Hackathon Roadmap & Scaling Vision

MEASURABLE IMPACT & OUTCOMES

- **75–80% Reduction in MTTI:** Cuts root-cause diagnosis from an industry average of 45+ minutes to under 90 seconds.
- **90% Alert Noise Elimination:** Condenses hundreds of cascading secondary alarms into a single coherent incident tree.
- **Sub-3-Minute Blast Containment:** Generates targeted containment actions before localized network faults spread across regions.
- **Direct Beneficiaries:** SREs avoid on-call panic; enterprises safeguard SLA guarantees; end-users experience uninterrupted digital services.
- **Cost Prevention:** Prevents multi-million-dollar outage penalties and reputational damage for cloud-native organizations.

24-HOUR HACKATHON MILESTONE ROADMAP

- **Hours 00–04 (Data Ingestion & Scenarios):** Define OpenTelemetry schemas, build streaming pipeline, and create Chaos failure generator simulating 4 outage patterns.
- **Hours 04–12 (Graph Engine & Causality):** Implement NetworkX dependency model, temporal anomaly parser, and root-cause scoring traversal algorithm.
- **Hours 12–18 (Dashboard & Containment UI):** Build Cytoscape.js interactive topology graph, chronological event timeline, and playbook generator.
- **Hours 18–24 (Testing & Pitch Polish):** Validate accuracy against chaos injection scenarios, measure MTTI benchmarks, and rehearse live demo.
- **Future Horizon:** Kernel-level eBPF continuous auto-instrumentation, AWS Systems Manager automation runner, and multi-cloud mesh support.

Conclusion & References

Deterministic Incident Investigation for Next-Generation Cloud Observability

SUMMARY & STRATEGIC TAKEAWAY

- **From Alert Chaos to Deterministic Precision:** TraceRoot replaces subjective, trial-and-error incident troubleshooting with mathematically sound causal graph traversal.
- **Explainable Proof Over Black-Box Guesses:** Engineers receive undeniable topological receipts linking anomalous logs to the initial point of failure, enabling confident decision-making.
- **Actionable Blast Containment:** Shifts the SRE paradigm from passive symptom monitoring to rapid root-cause isolation and immediate automated containment.

VERIFIABLE REFERENCES & INDUSTRY CITATIONS

- **Gartner IT Operations Study (2023):** The Cost of Enterprise Network Downtime and the Economics of SRE Incident Triage ($5,600 – $9,000/minute).
- **Google Site Reliability Engineering (SRE) Handbook:** Chapter 22: 'Addressing Cascading Failures' & Chapter 25: 'Managing Critical Alerts' (Beyer et al., O'Reilly Media).
- **AWS Well-Architected Framework:** Reliability Pillar: Best Practices for Resilient Cloud Architecture, Automated Failure Recovery & Incident Containment.
- **USENIX NSDI & ACM SIGCOMM:** Topology-Aware Root Cause Analysis in Large-Scale Cloud Microservices using Directed Causal Graphs.
- **OpenTelemetry & NetworkX Specifications:** Distributed Tracing Observability Data Models & Network Centrality Graph Traversal Standards.